

ให้

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$
$$C = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}, \quad a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}, \quad b = [3 \quad 0 \quad 8].$$

จงคำนวณผลคูณและผลบวกต่อไปนี้ หรือให้เหตุผลว่าทำไมถึงหาค่าไม่ได้ (Undefined)

1. $Ab^T + Bb^T, (A + B)b^T, bA, BA, B - B^T$
2. $A^T b, b^T B, (3A - 2B)^T a, a^T (3A - 2B)$
3. $ab - ba, -(4b)(7a), -28ba, 5abB$
4. $B^3, BC, (BC)^2, (BC)(BC)^T$
5. $IA, AI, A(I + A), A + IA$

คำตอบ

1. $\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}, \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}, \dots$

- 2.
- 3.
- 4.
- 5.

1.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$c = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

$$b = [3 \ 0 \ 8]$$

1.1)

$$Ab^T + Bb^T$$

$$b^T = \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$$

$$Ab^T = \begin{bmatrix} (18+0+8) \\ (30+0+4) \\ (-10+0+4) \end{bmatrix} = \begin{bmatrix} 26 \\ 34 \\ -6 \end{bmatrix}$$

3x1

$$Bb^T = \begin{bmatrix} (27+0+(-32)) \\ (12+0+0) \\ (-12+0+44) \end{bmatrix} = \begin{bmatrix} -5 \\ 12 \\ 32 \end{bmatrix}$$

$$Ab^T + Bb^T = \begin{bmatrix} 26 \\ 34 \\ -6 \end{bmatrix} + \begin{bmatrix} -5 \\ 12 \\ 32 \end{bmatrix} = \begin{bmatrix} 21 \\ 46 \\ 26 \end{bmatrix}$$

1.2) $(A+B)b^T$

$$A+B = \begin{bmatrix} 15 & 2 & -2 \\ 14 & 4 & 0 \\ -14 & 5 & 12 \end{bmatrix}$$

$$(A+B)b^T = \begin{bmatrix} 15 & 2 & -2 \\ 14 & 4 & 0 \\ -14 & 5 & 12 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 29 \\ 50 \\ 54 \end{bmatrix}$$

1.4) BA
3x3

$$= \begin{bmatrix} (54+40+40) & (-11-12-20) & (16+14-4) \\ (24+20+0) & (-2+(-21)+0) & (8+7+0) \\ (-24+0+(-110)) & (8+0+55) & (-8+0+77) \end{bmatrix}$$

$$BA = \begin{bmatrix} 134 & -43 & 26 \\ 44 & -23 & 15 \\ -134 & 63 & 69 \end{bmatrix}$$

1.3) bA

1x3

$$= \begin{bmatrix} (18+0+(-80)) & (-6+0+40) & (6+0+8) \end{bmatrix}$$

$$= \begin{bmatrix} -62 & 34 & 14 \end{bmatrix}$$

1.5) $B - B^T$

$$B^T = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$B - B^T = \begin{bmatrix} (9-9) & (4-4) & (-4-(-4)) \\ (4-4) & (7-7) & (0-0) \\ (-4-(-4)) & (0-0) & (11-11) \end{bmatrix}$$

$$B - B^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

2.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$c = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

$$b = [3 \quad 0 \quad 8]$$

2.1) $A^T b$

$$A^T = \begin{bmatrix} 6 & 10 & -10 \\ -2 & -3 & 5 \\ 2 & 1 & 1 \end{bmatrix}$$

$A^T b$ > ไม่สามารถหาได้ เพราะ ผลรวม แถวที่ 1 ไม่ลงตัวแถว แล้วมันคูณไม่พอ

2.2) $b^T B$

$$b^T = \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

$b^T B$: ไม่สามารถหาได้ เพราะ แถวที่ 1 ไม่ลงตัวแถว แล้วมันคูณไม่พอ

2.3) $(3A - 2B)^T a$

$$3A = \begin{bmatrix} 18 & -6 & 6 \\ 30 & -9 & 3 \\ -30 & 15 & 3 \end{bmatrix} \quad 2B = \begin{bmatrix} 18 & 8 & -8 \\ 8 & 14 & 0 \\ -8 & 0 & 22 \end{bmatrix}$$

$$(3A - 2B)^T = \begin{bmatrix} (18-18) & (-6-8) & (6-(-8)) \\ (30-8) & (-9-14) & (3-0) \\ (-30-(-8)) & (15-0) & (3-22) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -19 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 22 & -22 \\ -14 & -23 & 15 \\ 14 & 3 & -19 \end{bmatrix}$$

$$(3A - 2B)^T a$$

$$= \begin{bmatrix} 0 & 22 & -22 \\ -14 & -23 & 15 \\ 14 & 3 & -19 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 + 22 - 44 \\ -70 - 23 + 30 \\ 14 + 3 - 38 \end{bmatrix}$$

$$= \begin{bmatrix} -22 \\ -63 \\ 35 \end{bmatrix}$$

2.5

$$a^+ = [5 \ 1 \ 2]$$

$$(3A - 2B) = \begin{bmatrix} 0 & -14 & 14 \\ 22 & -23 & 3 \\ -22 & 15 & -14 \end{bmatrix}$$

min $\vec{v}$
2.4

$$= \begin{bmatrix} (0 + \cancel{22} - \cancel{44}) & (-14 - 23 + 30) & (14 + 3 - 18) \end{bmatrix}$$

$$= \begin{bmatrix} -22 & -6 & 35 \end{bmatrix}$$

3.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$c = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

$$b = [3 \quad 0 \quad 8]$$

$$3.1) ab - ba$$

$$ab = \begin{matrix} 3 \times 3 \\ \begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix} \end{matrix}$$

$$ba = \begin{matrix} 1 \times 1 \\ \begin{bmatrix} 15+0 & 16 \end{bmatrix} \end{matrix}$$

$$= \begin{bmatrix} 3 & 1 \end{bmatrix}$$

$$3.3) -28ba$$

$$= [-28 \times 31]$$

$$= [-868]$$

Ans ឃាត់ ឃាត់ ឃាត់ ឃាត់: ab រំលង 3×3 ba រំលង 1×1
 ទុកឃាត់ ឃាត់ ឃាត់ ឃាត់ ឃាត់ ឃាត់ ឃាត់ ឃាត់ ឃាត់ ឃាត់

$$3.2) -(4b)(7a)$$

$$= -(4 \times 7)(ba)$$

$$= -28(ba)$$

$$\text{ឃាត់ } 3.1 \quad ba = \begin{bmatrix} 3 & 1 \end{bmatrix}$$

$$-28ba = [-28 \times 31]$$

$$= [-868]$$

$$3.4) \quad 5abB$$

$$ab = \begin{bmatrix} 15 & 0 & 40 \\ 7 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix}$$

$$abB = \begin{bmatrix} (135+0-160) & (60+0+0) & (-60+0+440) \\ (27+0-32) & (12+0+0) & (-12+0+88) \\ (54+0-64) & (24+0+0) & (-24+0+176) \end{bmatrix}$$

$$= \begin{bmatrix} -25 & 60 & 380 \\ -5 & 12 & 76 \\ -10 & 24 & 152 \end{bmatrix}$$

$$5abB = \begin{bmatrix} -125 & 300 & 1900 \\ -25 & 60 & 380 \\ -50 & 120 & 760 \end{bmatrix} //$$

4.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$c = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

$$b = [3 \ 0 \ 8]$$

4.1) B^3

$$= B + B + B$$

$$B \times B = \begin{bmatrix} (81+16+16) & (36+28+0) & (-36+0-44) \\ (36+28+0) & (16+49+0) & (-16+0+0) \\ (-36+0-44) & (16+0+0) & (16+0+121) \end{bmatrix}$$

$$= \begin{bmatrix} 113 & 64 & -80 \\ 64 & 65 & -16 \\ -80 & -16 & 137 \end{bmatrix}$$

$$B^3 = \begin{bmatrix} 113 & 64 & -80 \\ 64 & 65 & -16 \\ -80 & -16 & 137 \end{bmatrix} \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} (1017+256+120) & (452+448+0) & (-452+0-880) \\ (576+260+64) & (256+455+0) & (-256+0-176) \\ (-720-64-548) & (-320-112+0) & (320+0+1507) \end{bmatrix}$$

$$= \begin{bmatrix} 1543 & 900 & -1332 \\ 900 & 711 & -432 \\ -1332 & -432 & 2827 \end{bmatrix} //$$

5.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$a = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

$$c = \begin{bmatrix} 3 & 1 \\ 0 & -2 \\ 4 & 0 \end{bmatrix}$$

$$b = [3 \ 0 \ 8]$$

5.1) IA

$$IA = A$$

Ans: IA គឺជា ធាតុត្រីកោណ បញ្ចេញនូវ កំរិត ១ ដូចជា ធាតុត្រីកោណ ធម្មតា

5.2) AA

Ans: ប្រើប្រាស់ ទិន្នន័យ 5.1

$$5.3 \quad A(I+A)$$

$$= AI + A^2$$

$$= A + A^2$$

$$A^2 = \begin{bmatrix} (36 - 20 - 20) & (-12 + 6 + 10) & (12 - 2 + 2) \\ (60 - 30 - 10) & (-20 + 4 + 5) & (20 - 3 + 1) \\ (-60 + 50 - 10) & (20 - 15 + 5) & (-20 + 5 + 1) \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 4 & 12 \\ 20 & -6 & 18 \\ -20 & 10 & -14 \end{bmatrix}$$

$$A + A^2 = \begin{bmatrix} (6 - 4) & (-2 + 4) & (2 + 12) \\ (10 + 20) & (-3 - 6) & (1 + 18) \\ (-10 - 20) & (5 + 10) & (1 - 14) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 14 \\ 30 & -9 & 19 \\ -30 & 15 & -13 \end{bmatrix}$$

5.4) $A + IA$

$$= A + A$$

$$= \begin{bmatrix} (6+6) & (-2-2) & (2+2) \\ (10+10) & (-3-3) & (1+1) \\ (-10-10) & (5+5) & (1+1) \end{bmatrix}$$

$$= \begin{bmatrix} 12 & -4 & 4 \\ 20 & -6 & 2 \\ -20 & 10 & 2 \end{bmatrix}$$